

Physical Chemistry Practical 2: Volumetric Analysis – Redox Titrations

Potassium Manganate(VII) Iron Analysis & Iodine–Thiosulfate Copper Determinations

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Physical Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Determination of Iron(II) in Mohr's Salt by Titration with Potassium Manganate(VII)

Cambridge Syllabus Reference: 9701/32/M/J/23/Q1 · [20 Marks]

Chemicals and Solutions Provided:

- FA 1 is $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .
- FA 2 is a solution prepared by dissolving 19.60 g dm^{-3} of hydrated ammonium iron(II) sulfate (Mohr's salt), $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$.
- FA 3 is 1.0 mol dm^{-3} sulfuric acid, H_2SO_4 .

Experimental Procedure:

- Fill the burette with $0.0200 \text{ mol dm}^{-3}$ KMnO_4 (FA 1). Note that readings should be taken at the upper meniscus or top edge due to the dark purple color.
- Pipette 25.0 cm^3 of FA 2 into a conical flask.
- Using a measuring cylinder, add 20 cm^3 of 1.0 mol dm^{-3} H_2SO_4 (FA 3) to acidify the solution.
- Titrate with FA 1 until a permanent faint pale pink color persists for at least 30 seconds (no external indicator is required: KMnO_4 is self-indicating).
- Record your readings and obtain concordant titres within 0.10 cm^3 .

(a) Record your burette readings and titre volumes for the titration of FA 2 with FA 1 in the table below. [5]

Burette Readings	Rough	Titration 1	Titration 2	Titration 3
Final burette reading / cm^3				
Initial burette reading / cm^3				
Titre / cm^3				
Best titration results ()				

(b) Calculate your mean titre of FA 1. [1]

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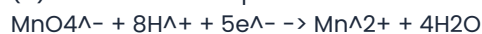
(c) Calculate the amount, in moles, of manganate(VII) ions, MnO_4^- , present in your mean titre of FA 1. [2]

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(d) The ionic half-equations are:



Write the overall balanced ionic equation, and deduce the amount, in moles, of Fe^{2+} present in 25.0 cm^3 of FA 2. [3]

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(e) Calculate the concentration of Fe^{2+} in FA 2 in mol dm^{-3} , and hence calculate the relative formula mass (Mr) of the Mohr's salt sample (which contains 19.60 g dm^{-3} of the hydrated salt). [3]

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(f) The formula of Mohr's salt is $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$. Given that the relative formula mass of anhydrous $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$ is 284.0, calculate the value of x (the number of water of crystallization molecules). [Ar: H = 1.0, O = 16.0] [3]

(g) Explain why hydrochloric acid (HCl) cannot be used instead of sulfuric acid (H_2SO_4) to acidify the reaction mixture in this titration. [3]

Question 2: Iodine–Thiosulfate Titration: Determination of Copper in an Alloy

Cambridge Syllabus Reference: 9701/34/M/J/23/Q1 · [20 Marks]

Chemicals and Solutions Provided:

- FA 4 is an aqueous solution of copper(II) sulfate, CuSO_4 , obtained by dissolving 3.50 g of a copper brass alloy in nitric acid and diluting to 250 cm^3 .
- FA 5 is $0.100 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.
- Aqueous potassium iodide, KI (approx 0.5 mol dm^{-3}).
- Starch indicator solution.

Experimental Procedure:

1. Pipette 25.0 cm^3 of FA 4 into a conical flask.
2. Using a measuring cylinder, add 15 cm^3 of aqueous potassium iodide. A thick brown precipitate/suspension of iodine and copper(I) iodide forms: $2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI(s)} + \text{I}_2(\text{aq})$.
3. Titrate the liberated iodine with $0.100 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$ (FA 5) from the burette until the dark brown color fades to a pale straw-yellow.
4. Add 1 cm^3 of starch indicator. The solution turns dark blue-black.
5. Continue adding FA 5 dropwise with vigorous swirling until the blue-black color disappears, leaving a milky-white precipitate of CuI . Record the endpoint volume.

(a) Record your titration results for the titration of the liberated iodine with FA 5 in the table below. [5]

Burette Readings	Rough	Titration 1	Titration 2	Titration 3
Final burette reading / cm^3				
Initial burette reading / cm^3				
Titre / cm^3				
Best titration results ()				

(b) Calculate your mean titre of FA 5. [1]

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(c) Calculate the amount, in moles, of thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$, present in your mean titre of FA 5. [2]

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(d) The titration reaction is: $2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$.

Calculate the amount, in moles, of I_2 liberated in the conical flask. [2]

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(e) Given that 2 moles of Cu^{2+} liberate 1 mole of I_2 ($2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI} + \text{I}_2$), calculate the amount, in moles, of copper present in the 25.0 cm^3 sample of FA 4, and hence in the original 250 cm^3 flask. [3]

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(f) Calculate the mass of copper in the alloy sample, and determine the percentage by mass of copper in the alloy.

[Ar: Cu = 63.5; Sample mass = 3.50 g] [3]

(g) Why must the starch indicator NOT be added at the beginning of the titration when the iodine concentration is high? [2]

(h) State the function of adding potassium thiocyanate (KSCN) just before the endpoint in precision copper determinations. [2]

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical 2: Volumetric Analysis – Redox Titrations

Question 1: Determination of Iron(II) in Mohr's Salt by Titration with Potassium Manganate(VII) [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly, readings to 0.05 cm^3 , concordancy $\leq 0.10 \text{ cm}^3$, accuracy marks [5].	[5]
(b)	Mean titre correctly calculated to 2 decimal places (e.g. 25.00 cm^3).	[1]
(c)	Moles of $\text{MnO}_4^- = 0.0200 \text{ mol dm}^{-3} \times (25.00 / 1000) = 5.00 \times 10^{-4} \text{ mol}$. [2]	[2]
(d)	Overall equation: $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$ [1] Mole ratio is 1 MnO_4^- : 5 Fe^{2+} . Moles of Fe^{2+} in $25.0 \text{ cm}^3 = 5 \times (5.00 \times 10^{-4}) = 2.50 \times 10^{-3} \text{ mol}$ [2]	[3]
(e)	Concentration of $\text{Fe}^{2+} = 2.50 \times 10^{-3} \times (1000 / 25.0) = 0.100 \text{ mol dm}^{-3}$ [1] Mr of Mohr's salt = Mass concentration / Molar concentration = $19.60 \text{ g dm}^{-3} / 0.100 \text{ mol dm}^{-3} = 196.0 \text{ g/mol}$ (anhydrous) or 392.0 g/mol for hexahydrate [2]	[3]
(f)	Mass of water per mole = $\text{Mr}(\text{total}) - \text{Mr}(\text{anhydrous}) = 392.0 - 284.0 = 108.0 \text{ g}$ [1] $x = 108.0 / 18.0 = 6.0$ [1] Therefore, $x = 6$ (Mohr's salt is a hexahydrate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$) [1]	[3]
(g)	Potassium manganate(VII) is a powerful oxidizing agent that would oxidize chloride ions (Cl^-) in HCl to chlorine gas (Cl_2): $2\text{MnO}_4^- + 10\text{Cl}^- + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 5\text{Cl}_2 + 8\text{H}_2\text{O}$. [1] This would consume extra KMnO_4 , leading to an artificially high titre volume and causing false calculated values for iron. Sulfuric acid contains sulfur in its maximum oxidation state (+6) and cannot be oxidized. [2]	[3]

Question 2: Iodine–Thiosulfate Titration: Determination of Copper in an Alloy [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Table formatted correctly, readings to 0.05 cm^3 , concordancy $\leq 0.10 \text{ cm}^3$, accuracy marks [5].	[5]
(b)	Mean titre calculated to 2 decimal places (e.g. 22.50 cm^3).	[1]
(c)	Moles of $\text{S}_2\text{O}_3^{2-} = 0.100 \times (22.50 / 1000) = 2.25 \times 10^{-3} \text{ mol}$. [2]	[2]
(d)	Moles of $\text{I}_2 = \text{Moles of } \text{S}_2\text{O}_3^{2-} / 2 = 2.25 \times 10^{-3} / 2 = 1.125 \times 10^{-3} \text{ mol}$. [2]	[2]
(e)	Moles of Cu^{2+} in $25.0 \text{ cm}^3 = 2 \times \text{Moles of } \text{I}_2 = 2 \times 1.125 \times 10^{-3} = 2.25 \times 10^{-3} \text{ mol}$ [1] Total moles of Cu in $250 \text{ cm}^3 = 2.25 \times 10^{-3} \times 10 = 2.25 \times 10^{-2} \text{ mol}$ (0.0225 mol) [2]	[3]
(f)	Mass of copper = $0.0225 \text{ mol} \times 63.5 \text{ g/mol} = 1.429 \text{ g}$ [1] % copper in alloy = $(1.429 \text{ g} / 3.50 \text{ g}) \times 100\% = 40.8\%$ [2]	[3]
(g)	At high iodine concentrations, iodine binds irreversibly to the starch helical amylose coils, preventing rapid dissociation at the endpoint and leading to an inaccurate, delayed endpoint. Starch is added only when the straw-yellow color indicates most iodine has reacted. [2]	[2]
(h)	Copper(I) iodide (CuI) adsorbs iodine molecules onto its surface. Adding KSCN displaces adsorbed iodine (forming CuSCN precipitate), releasing all iodine into solution for complete titration with thiosulfate, giving a sharper and more accurate endpoint. [2]	[2]